

From Conversation to Delegation: Multi-Layer Network Analysis of Agentic AI Sessions

Alexander Towell*

Department of Computer Science
Southern Illinois University Edwardsville
Edwardsville, IL, USA

John Matta

Department of Computer Science
Southern Illinois University Edwardsville
Edwardsville, IL, USA

Abstract

How does the cognitive trace of AI-assisted work change when the AI shifts from passive interlocutor to autonomous agent? Prior work in this series established a *cognitive MRI* methodology for analyzing conversational AI archives as semantic similarity networks, revealing super-linear densification ($\gamma = 1.405$), sub-linear preferential attachment, and stable community structure in a single user’s ChatGPT history. Here we extend this framework to *agentic* AI by analyzing 3,945 Claude Code sessions (1,061 human-initiated parents and 2,884 AI-spawned subagents) generated by the same human subject over 101 days. Agentic sessions create two simultaneous network layers: a *semantic similarity layer* encoding topical relationships and a *delegation hierarchy* encoding parent–child task structure. We find that agentic work intensifies densification ($\gamma = 1.707$ vs. 1.405, both $R^2 > 0.98$) while weakening modularity (0.391 vs. 0.750), reflecting a shift from topic-driven to project-driven knowledge organization. The delegation network exhibits heavy-tailed fan-out (mean 2.72, max 156) with distinct structural roles for different agent types. Critically, the two layers are weakly coupled: inter-layer degree correlation is low ($\rho = 0.115$, $p < 10^{-13}$) and community assignments overlap only moderately (NMI = 0.581). Semantic centrality does not predict delegation behavior, and vice versa. This complementarity demonstrates that single-layer analysis—the approach of our prior work—systematically misses the dimension that multiplex analysis reveals. As AI transitions from conversational partner to autonomous agent, the cognitive traces it produces become structurally richer and demand correspondingly richer analytical frameworks.

Author summary

When you ask a chatbot a question, it answers and you move on. When you use an AI coding assistant like Claude Code, something different happens: the AI reads your files, writes code, runs tests, and spawns child agents that work autonomously on subtasks. This creates two kinds of structure in the resulting archive. First, sessions that cover similar topics are semantically

*Corresponding author. Email: atowell@siue.edu. ORCID: 0000-0001-6443-9897

connected—the same kind of structure we found in our earlier studies of ChatGPT conversations. Second, parent sessions are linked to the child agents they spawned, creating a delegation hierarchy. We analyzed nearly 4,000 Claude Code sessions and found that these two structures are largely independent: knowing how semantically central a session is tells you almost nothing about how many child agents it spawned, and vice versa. Looking at only one layer misses what the other reveals. This matters because agentic AI is rapidly becoming the dominant mode of human–AI interaction, and understanding the cognitive traces it leaves requires tools that can capture both what work is about and how it is organized.

1 Introduction

Conversational AI has created a new medium for knowledge exploration. Over the past three years, millions of users have generated conversation archives that constitute externalized records of cognitive activity [Zhao et al., 2023]. In prior work, we introduced a *cognitive MRI* methodology that transforms these archives into semantic similarity networks, where conversations become nodes and edges connect semantically related pairs [Towell and Matta, 2025] (hereafter *Paper 1*). Analyzing a single user’s ChatGPT archive, we identified 15 knowledge communities, heterogeneous network topology, and bridge conversations facilitating cross-domain knowledge transfer. A temporal extension [Towell and Matta, 2026b] (hereafter *Paper 2*) revealed that this network exhibits super-linear densification ($\gamma = 1.405$, $R^2 = 0.993$), sub-linear preferential attachment ($\beta = 0.763$), and early community stabilization over 29 months of growth—evolution patterns characteristic of real-world complex networks [Leskovec et al., 2007, Barabási and Albert, 1999].

But conversational AI is passive. The human initiates every turn, the AI responds, and the conversation proceeds as a single linear thread. The cognitive trace is correspondingly simple: a flat sequence of exchanges whose only structure is topical similarity.

Agentic AI changes this dynamic fundamentally. Tools such as Claude Code, GitHub Copilot, and Cursor do not merely answer questions—they read files, write code, execute commands, and spawn child agents that work autonomously on delegated subtasks [Wang et al., 2024, Xi et al., 2023]. A single human-initiated session may produce a tree of subagent sessions, each performing specialized work: one agent refactors a module, another writes tests, a third updates documentation. The result is not a linear conversation but a hierarchical delegation structure, a qualitatively different kind of cognitive trace.

This hierarchical structure creates a natural *multiplex network* [Boccaletti et al., 2014, Kivela et al., 2014]. The semantic similarity layer, constructed the same way as in our prior work, captures *what* sessions are about: their topical relationships. The delegation layer, derived from parent–child metadata, captures *who did what for whom*: the structural organization of work. Neither layer alone tells the full story. Semantic analysis reveals topical coherence but misses how labor was distributed. Delegation analysis reveals task structure but misses the topical connections that link unrelated branches of the hierarchy.

The central question of this paper is whether these two layers encode the same structure or reveal complementary dimensions of cognitive activity.

We address this question through four research questions:

RQ1 (Semantic): Do agentic knowledge networks exhibit the same macroscopic laws as conversational ones?

RQ2 (Delegation): What network properties characterize AI agent delegation structure?

RQ3 (Temporal): Does the agentic network densify at daily resolution?

RQ4 (Multi-layer): Do semantic and delegation layers encode the same or different organization?

RQ5 (Validation): Are the observed network properties genuine or artifacts of the degree sequence?

Our analysis of 3,945 Claude Code sessions from the same human subject studied in Papers 1–2 reveals three principal findings. First, agentic work *intensifies* densification ($\gamma = 1.707$ vs. 1.405) while *weakening* modularity (0.391 vs. 0.750), reflecting a structural shift from topic-driven to project-driven knowledge organization. Second, the delegation network exhibits heavy-tailed fan-out and distinct structural roles for different agent types. Third—and most critically—the semantic and delegation layers are *weakly coupled*: inter-layer degree correlation is low ($\rho = 0.115$), and community structure overlaps only moderately (NMI = 0.581). This means that single-layer analysis, the approach of our prior work, systematically misses the complementary dimension that the multiplex view reveals.

These findings contribute to multiplex network theory by providing an empirical case in which two naturally arising layers encode demonstrably different organizational structure, and to the emerging study of human–AI interaction by showing that agentic AI produces cognitive traces that are structurally richer than those of conversational AI and require correspondingly richer analytical frameworks.

Section 2 reviews related work on multiplex networks, densification laws, agentic AI, and the cognitive MRI framework. Section 3 describes the dataset, network construction, and analysis methods. Section 4 presents results organized by research question. Section 5 interprets the findings and identifies limitations. Section 6 concludes.

2 Related Work

2.1 Multiplex and multi-layer networks

Real-world systems often exhibit multiple types of relationships among the same set of entities. Multiplex network theory provides the mathematical framework for analyzing such systems. Boccaletti et al. [2014] offer a comprehensive review, covering structural measures, dynamical processes, and applications across social, biological, and technological domains. De Domenico et al. [2013] formalize

the tensor representation of multilayer networks, enabling rigorous definition of inter-layer dependencies. Kivelä et al. [2014] survey the broader landscape of multilayer network types, distinguishing multiplex networks (same nodes, different edge types) from other multilayer configurations.

A central insight from this literature is that analyzing layers in isolation can miss structure that emerges from their interaction. Battiston et al. [2014] introduce the *multiplex participation coefficient*, quantifying the extent to which a node’s connectivity is distributed across layers. Nodes with high participation are active in multiple relational contexts; those with low participation are specialized to one layer. Szell et al. [2010] demonstrate this principle empirically in a massively multiplayer online game, showing that positive (friendship), negative (enmity), and economic (trading) layers encode largely distinct social structure. Bullmore and Sporns [2009] find analogous complementarity in brain networks, where structural connectivity (white matter tracts) and functional connectivity (correlated activation) capture different aspects of neural organization.

The multiplex participation coefficient and inter-layer degree correlation are the primary tools we use to quantify layer independence in our agentic AI network.

2.2 Densification laws and temporal network evolution

Leskovec et al. [2005] discovered that many real-world networks exhibit *densification*: the number of edges grows super-linearly with the number of nodes, following a power law $e(t) \propto n(t)^\gamma$ with $\gamma > 1$. This pattern appears across citation networks ($\gamma = 1.69$), patent networks ($\gamma = 1.26$), and autonomous systems graphs ($\gamma = 1.18$) [Leskovec et al., 2007]. Densification implies that as a network grows, its members increasingly find connections to existing content rather than remaining isolated.

Temporal network analysis encompasses the study of how network structure evolves [Holme and Saramäki, 2012]. Barabási and Albert [1999] showed that preferential attachment—new nodes connecting preferentially to well-connected existing nodes—produces scale-free degree distributions. Mucha et al. [2010] extended community detection to time-dependent and multiplex networks via generalized modularity optimization.

Our previous work [Towell and Matta, 2026b] demonstrated that a single user’s ChatGPT conversation archive densifies with $\gamma = 1.405$ ($R^2 = 0.993$) over 29 monthly snapshots. The present work tests whether this pattern persists, intensifies, or attenuates in the agentic setting, where sessions are shorter, more numerous, and often specialized subagent tasks rather than broad conversational explorations.

2.3 Agentic AI and tool use

The emergence of agentic coding assistants represents a qualitative shift in human–AI interaction. Unlike conversational AI, which responds to user prompts within a single dialogue thread, agentic systems autonomously read files, write code, execute commands, and delegate subtasks to child agents [Wang et al., 2024, Xi et al., 2023]. This delegation pattern—a parent task spawning child

tasks that may themselves spawn further children—creates tree-structured workflows reminiscent of task decomposition in distributed computing [Wooldridge, 2009, Dorri et al., 2018].

Shen et al. [2024] describe a paradigm in which a controller LLM decomposes complex tasks and delegates subtasks to specialized models, organizing work through hierarchical planning. Such systems produce explicit delegation traces—parent–child relationships between sessions—that are absent in conversational AI. These traces constitute a natural directed network layer whose properties have not, to our knowledge, been analyzed from a complex networks perspective.

The multi-agent systems literature [Dorri et al., 2018] provides relevant concepts: task allocation, coordination protocols, and emergent collective behavior. However, that literature typically studies systems of peer agents with predefined roles, whereas agentic AI delegation is dynamic, with agents spawned on demand for specific subtasks and dissolved upon completion.

2.4 The cognitive MRI framework

This paper is the third in a series applying complex network analysis to personal AI archives. Paper 1 [Towell and Matta, 2025] introduced the *cognitive MRI* methodology: embedding conversations into a semantic space, computing pairwise cosine similarity, thresholding to construct a network, and analyzing the resulting graph’s community structure, centrality distribution, and bridge conversations. Applied to the giant component of a ChatGPT archive (449 conversations at similarity threshold $\theta = 0.9$), it revealed 15 knowledge communities and three bridge conversation types.

Paper 2 [Towell and Matta, 2026b] extended this to temporal analysis, tracking network evolution over 29 months. The key findings—super-linear densification ($\gamma = 1.405$), sub-linear preferential attachment ($\beta = 0.763$), and early community stabilization—established that personal knowledge exploration self-organizes according to the same macroscopic laws observed in collective knowledge systems such as citation networks [Leskovec et al., 2007, de Solla Price, 1965].

The present work extends the framework in two directions: from conversational to agentic AI, and from single-layer to multiplex analysis. The theoretical framing draws on distributed cognition [Hutchins, 1995] and the extended mind thesis [Clark and Chalmers, 1998]. Hutchins argued that cognitive processes are distributed across people, artifacts, and environments; Clark and Chalmers extended this to argue that external resources functioning in the right way constitute genuine parts of a cognitive system. The agentic AI archive is precisely such a resource: a structured record of cognitive labor distribution whose network analysis reveals patterns invisible in the raw session log.

3 Methods

3.1 Dataset

The dataset comprises session logs from Claude Code, an agentic AI coding assistant, generated by one of the authors between November 24, 2025 and March 4, 2026 (101 calendar days, 49 with at

Table 1: Dataset comparison: conversational (ChatGPT) versus agentic (Claude Code). ^cText-only content used for embedding; tool-use and tool-result blocks stripped during preprocessing.

Property	ChatGPT (Paper 2)	Claude Code (this work)
Sessions	1,908	3,945
Messages	—	374,376
Observation period	29 months	101 days
Temporal resolution	Monthly	Daily
Session hierarchy	Flat	Parent-child (depth ≤ 2)
Content modality	Text only	Text, tool use, tool result ^c
Connected at $\theta = 0.9$	601	3,068
Isolated at $\theta = 0.9$	1,307	877
Same human subject	Yes	Yes

least one session). This is the same human subject studied in Papers 1–2 [Towell and Matta, 2025, 2026b], enabling direct comparison across AI modalities.

Sessions were extracted from the local memex SQLite database maintained by Claude Code. Each session record includes a unique identifier, creation timestamp, message sequence, and—for subagent sessions—a `parent_conversation_id` linking it to the session that spawned it. The dataset contains:

- 1,061 *parent sessions*: human-initiated top-level sessions.
- 2,884 *subagent sessions*: AI-spawned child sessions created when a parent (or the system) delegates a subtask.
- 3,945 total sessions containing 374,376 messages.

Subagent sessions fall into three categories based on identifier patterns:

- *User-spawned* (2,077; 72.0%): substantive delegated work, including code writing, testing, and analysis tasks. These range from brief lookups to extended sessions with up to 1,057 messages.
- *Compact* (518; 18.0%): context window compaction sessions triggered when the parent session’s context exceeds capacity.
- *Prompt suggestion* (289; 10.0%): auto-generated follow-up suggestions.

Table 1 compares the agentic dataset with the conversational dataset from Paper 2.

The agentic dataset is substantially denser in time (3,945 sessions over 101 days vs. 1,908 over 29 months) and richer in structure (hierarchical delegation, mixed content modalities). Sessions were generated through authentic software development, research, and writing tasks with no anticipation of future network analysis.

3.2 Semantic layer construction

The semantic layer is an undirected weighted graph in which nodes are sessions and edges connect semantically similar pairs. Construction follows the pipeline established in Papers 1–2 for maximal comparability:

1. **Content preprocessing.** We extract text-only content from each session, stripping tool-use blocks, tool-result blocks, and thinking blocks. This ensures comparability with the ChatGPT methodology, where conversations contain only text. We note that this discards information characteristic of agentic sessions (code, file contents, command output); the implications are discussed in Section 5.6.
2. **Embedding.** Each session’s text content is embedded using `nomic-embed-text` [Nussbaum et al., 2024] via Ollama, the same model used in Papers 1–2. For sessions exceeding the model’s 8,192-token context window, content is split into 256-word chunks, each chunk is embedded independently, and the chunk embeddings are averaged. The resulting 768-dimensional embedding vector is L2-normalized.
3. **Pairwise similarity.** Cosine similarity is computed for all $\binom{3,945}{2} = 7,779,540$ session pairs.
4. **Thresholding.** Edges are retained for pairs with cosine similarity $\geq \theta = 0.9$. This threshold was validated by a 63-configuration ablation study in Paper 1 [Towell and Matta, 2025], which identified $\theta = 0.9$ as the phase transition point balancing connectivity with specificity. We apply it without independent re-validation for the agentic data; this limitation is discussed in Section 5.6.

The resulting semantic layer contains 3,945 nodes and 287,693 undirected edges.

3.3 Delegation layer construction

The delegation layer is a directed graph in which each edge points from a parent session to a child session it spawned. Unlike the semantic layer, which requires embedding, similarity computation, and thresholding, the delegation layer is deterministic: edges are derived directly from the `parent_conversation_id` metadata field.

The delegation graph contains 3,945 nodes and 2,884 directed edges. It is a forest (a collection of rooted trees): each parent session is a root, its subagent sessions are children, and the maximum depth is 2 (parent $\rightarrow$ child; no grandchild sessions were observed). Of the 1,061 parent sessions, 365 spawned at least one child. The mean fan-out across all parents (including those with zero children) is 2.72; among parents with at least one child, the mean is 7.90 and the maximum is 156.

3.4 Temporal snapshot construction

We construct cumulative daily snapshots to track network evolution. For each day t in the observation period, the snapshot $G(t) = (V(t), E(t))$ contains all sessions created on or before day t and all

semantic edges between them:

$$V(t) = \{v \in V \mid \text{created}(v) \leq t\} \quad (1)$$

$$E(t) = \{(u, v) \in E_{\text{semantic}} \mid u \in V(t) \wedge v \in V(t)\} \quad (2)$$

This produces 49 daily snapshots (one per active day) spanning November 24, 2025 through March 4, 2026. The cumulative construction models the irreversibility of completed work: once a session exists, it permanently enriches the knowledge landscape. This mirrors the monthly snapshot construction in Paper 2 but at higher temporal resolution.

For each snapshot, we compute node count, edge count, density, number of connected components, giant component size and fraction, number of parent sessions, number of child sessions, and mean fan-out.

3.5 Network metrics

We compute the following metrics on the final-snapshot networks.

Semantic layer. Density, number of connected components, giant component size and fraction, average and maximum degree, average clustering coefficient [Newman, 2003], transitivity, average shortest path length (within the giant component), Louvain modularity and community count [Blondel et al., 2008] (random seed 42 for reproducibility), and degree assortativity. All network computations use NetworkX [Hagberg et al., 2008].

Delegation layer. Node and edge counts, fan-out distribution (number of children per parent), delegation ratio (total messages across all children of a parent divided by the parent’s message count), and agent type breakdown.

Temporal. Densification law fitting following Leskovec et al. [2005]: OLS regression on $\log e(t)$ versus $\log n(t)$, yielding exponent γ , coefficient of determination R^2 , and p -value. We exclude the initial single-node snapshot (which has no edges) from the fit. We note that OLS on log-transformed data is the standard approach for densification [Leskovec et al., 2007]; formal power-law testing [Clauset et al., 2009] could provide additional rigor but is beyond our scope.

Multi-layer. Three metrics quantify the relationship between semantic and delegation layers:

1. *Inter-layer degree correlation* (Spearman’s ρ): correlation between each node’s semantic degree (number of edges in the semantic layer) and delegation degree (in-degree plus out-degree in the delegation layer). A high $|\rho|$ would indicate that semantically central nodes are also delegation hubs; a low $|\rho|$ indicates the layers encode different structure.

2. *Multiplex participation coefficient* [Battiston et al., 2014]: for each node i with degree k_i^α in layer α ,

$$P_i = 1 - \sum_{\alpha=1}^M \left(\frac{k_i^\alpha}{o_i} \right)^2 \quad (3)$$

where $o_i = \sum_{\alpha} k_i^\alpha$ is the overlapping degree and $M = 2$ is the number of layers. This is the unnormalized Herfindahl-based form; for $M = 2$, it ranges from $P_i = 0$ (all connections in one layer) to $P_i = 0.5$ (equally distributed).

3. *Community structure comparison* (Normalized Mutual Information, NMI): we independently detect communities in each layer and compare the resulting partitions using NMI [Danon et al., 2005]. For the semantic layer, we use Louvain community detection. For the delegation layer, connected components of the undirected forest provide a natural partition: each tree represents a project and its subtasks, making components the appropriate organizational unit. NMI = 1 indicates identical partitions; NMI = 0 indicates independence.

3.6 Comparison framework

A key feature of our experimental design is that the agentic and conversational datasets share the same human subject, the same embedding model (`nomic-embed-text`), and the same similarity threshold ($\theta = 0.9$). This controls for individual differences, embedding quality, and threshold effects, providing a descriptive comparison that holds several key variables constant. However, confounds remain: the two datasets use different AI platforms, span different time periods (29 months vs. 101 days), and differ in content structure (free-form conversation vs. tool-mediated coding sessions). We interpret cross-dataset comparisons as suggestive rather than causal.

Metrics are compared directly between the two datasets where methodologically equivalent. We note that the observation periods differ substantially (29 months vs. 101 days), which affects temporal comparisons; this is addressed in the discussion.

4 Results

4.1 Semantic network structure (RQ1)

Table 2 presents a side-by-side comparison of semantic network metrics between the conversational (ChatGPT) and agentic (Claude Code) datasets.

Several patterns distinguish the agentic semantic network from its conversational counterpart (Table 2).

Intensified densification. The densification exponent increases from $\gamma = 1.405$ in the conversational network to $\gamma = 1.707$ in the agentic network (both $R^2 > 0.98$; Fig. 2). For every doubling of the connected node count, the conversational network adds a factor of $2^{1.405} \approx 2.65$ edges; the

Table 2: Semantic network metrics: conversational (ChatGPT) versus agentic (Claude Code). Both networks use the same embedding model, similarity threshold ($\theta = 0.9$), and human subject. All metrics computed on the full network including isolated nodes. ^aSessions with ≥ 1 edge. ^bFraction of total sessions.

Metric	ChatGPT (Paper 2)	Claude Code (this work)
Total sessions	1,908	3,945
Connected sessions ^a	601	3,068
Edges ($\theta = 0.9$)	1,718	287,693
Density	0.016	0.037
Giant component size	449	2,465
Giant component fraction ^b	23.5%	62.5%
Average degree	7.19	145.85
Max degree	—	843
Clustering coefficient	0.60	0.48
Transitivity	—	0.957
Avg. shortest path (GC)	—	4.009
Modularity (Louvain)	0.750	0.391
Communities	15	1,084
Degree assortativity	−0.13	+0.90
Densification γ	1.405	1.707

253 agentic network adds $2^{1.707} \approx 3.26$ edges. Agentic sessions create cross-domain semantic connections
254 more aggressively than conversational ones.

255 **Reduced modularity.** Modularity drops from 0.750 to 0.391, and the number of communities
256 increases from 15 to 1,084. The conversational network organized into a small number of well-
257 separated knowledge communities (machine learning, statistics, philosophy, programming, etc.). The
258 agentic network contains many small, less distinct clusters. This reflects a shift from topic-driven to
259 project-driven organization: a single agentic session may involve debugging, testing, documentation,
260 and deployment—spanning multiple traditional topic categories—whereas a conversation typically
261 remains within one domain.

262 **High assortativity.** Degree assortativity reverses from −0.13 (mildly disassortative) to +0.90
263 (strongly assortative). In the agentic network, well-connected sessions preferentially link to other
264 well-connected sessions, forming a “rich club” of complex, multi-topic sessions. This is consistent
265 with the project-driven structure: large, complex sessions that touch many domains are semantically
266 similar to each other precisely because they share this multi-domain character.

267 **Higher giant component fraction.** The giant component contains 62.5% of all sessions, up
268 from 23.5% in the ChatGPT archive (449 of 1,908 total sessions; Papers 1–2 reported 74.7% among
269 the 601 connected sessions only). This higher connectivity reflects the multi-topic nature of agentic
270 sessions: project-driven work spanning multiple domains creates more cross-topic similarity links.

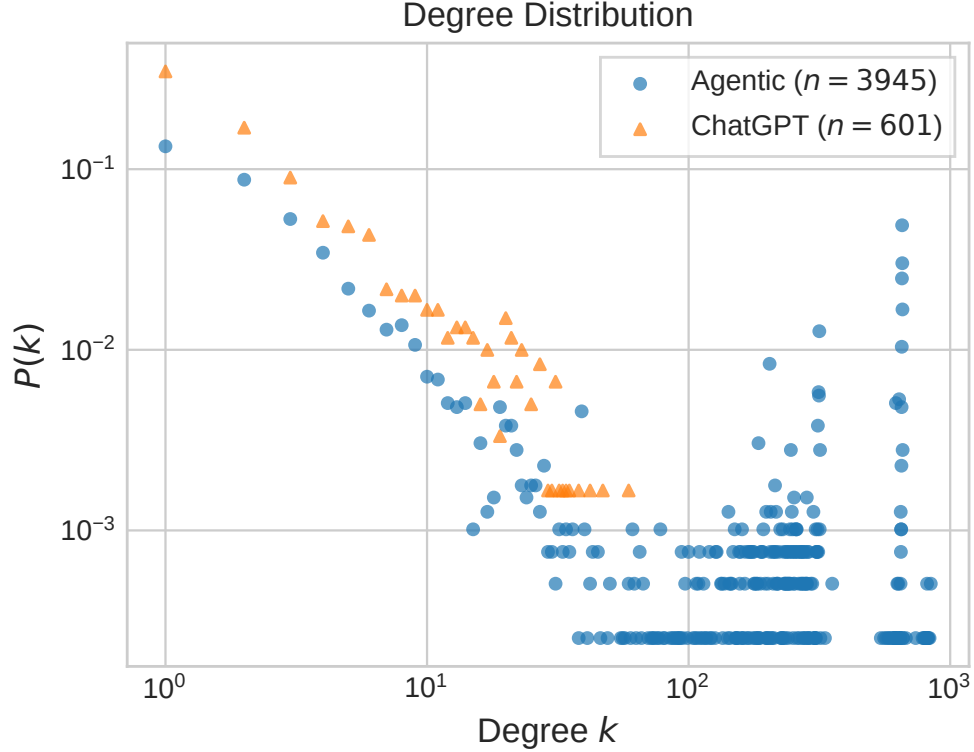


Figure 1: Degree distribution comparison (log-log). The agentic network exhibits a broader, heavier-tailed distribution centered at much higher degrees than the conversational network.

Nevertheless, the network fragments into 1,070 connected components, reflecting the large number of brief, narrowly focused subagent sessions (877 isolated nodes) whose content is too specialized to reach the $\theta = 0.9$ similarity threshold with the broader network.

Higher density and average degree. Density increases from 0.016 to 0.037 and average degree from 7.19 to 145.85. The agentic network is substantially more interconnected, with the typical session sharing semantic similarity (≥ 0.9) with nearly 146 other sessions.

Clustering–transitivity gap. The average clustering coefficient (0.48) and global transitivity (0.957) exhibit an unusually large gap. This reflects the network’s degree heterogeneity: the dense rich-club of high-degree nodes closes nearly all triangles among themselves (driving transitivity up), while the many low-degree peripheral nodes have sparser local neighborhoods (pulling average clustering down). This gap is consistent with the strong positive assortativity (+0.90). The degree distribution (Fig. 1) illustrates the heavy-tailed, high-degree structure of the agentic network compared to the conversational one.

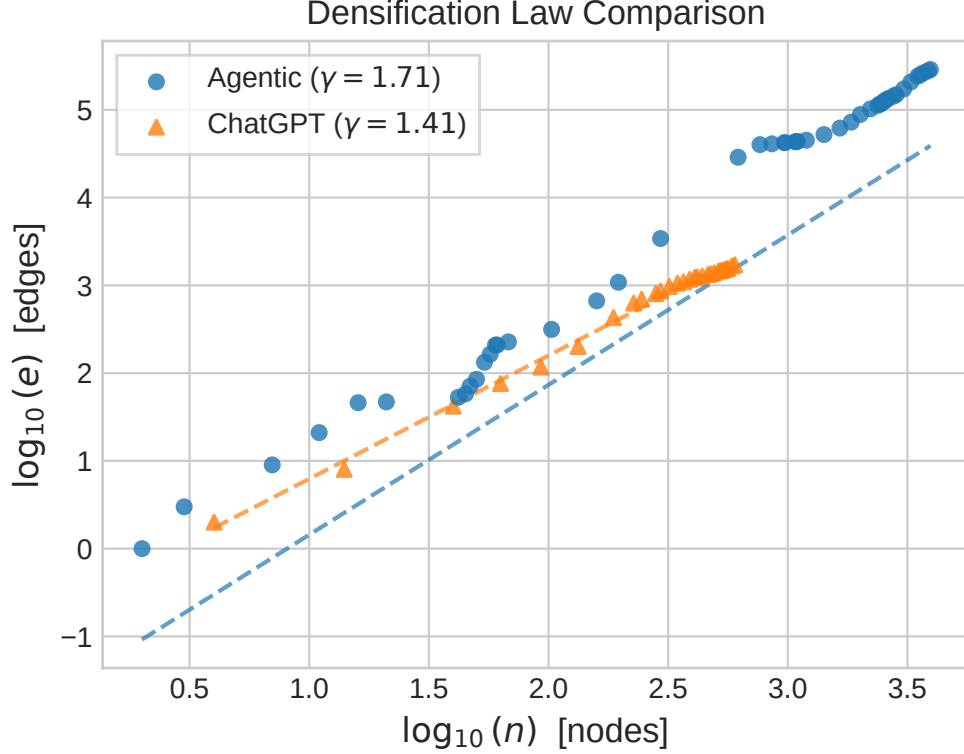


Figure 2: Densification law comparison. Both datasets follow power laws ($e \propto n^\gamma$), but the agentic network densifies more aggressively ($\gamma = 1.71$ vs. 1.41).

4.2 Delegation network structure (RQ2)

The delegation layer consists of 3,945 nodes and 2,884 directed edges (parent $\rightarrow$ child). Table 3 summarizes its properties.

Heavy-tailed fan-out. The fan-out distribution (Fig. 3) is heavy-tailed: most parent sessions spawn 0–3 subagents, but a long tail extends to 156. The session with 156 subagents represents an intensive multi-day development effort in which the parent repeatedly delegated subtasks as new requirements emerged. This distribution is consistent with the heavy-tailed task complexity distributions observed in software development generally.

Delegation ratio. The delegation ratio—the total messages across all children of a parent divided by the parent’s message count—has a mean of 0.316 (Fig. 4), indicating that a parent’s collective subagent work typically produces about one-third the message volume of the parent itself. However, the maximum of 7.0 shows that some parents’ children collectively produce seven times the parent’s message count, suggesting cases where delegated work substantially exceeded the initiating session in scope.

Table 3: Delegation network summary. ^aPercentages are of the 2,884 subagent sessions.

Property	Value
Nodes	3,945
Directed edges	2,884
Agent types ^a : user-spawned	2,077 (72.0%)
compact	518 (18.0%)
prompt suggestion	289 (10.0%)
Fan-out (mean)	2.72
Fan-out (max)	156
Delegation ratio (mean)	0.316
Delegation ratio (max)	7.0
Maximum tree depth	2

Agent type roles. The three agent types play distinct structural roles. User-spawned agents (72.0%) represent genuine delegated intellectual labor—the parent identifies a subtask and creates an agent to handle it. Compact agents (18.0%) are infrastructure: they are triggered automatically when a parent session’s context window fills, compressing the conversation to free space. Prompt suggestion agents (10.0%) are auto-generated follow-up suggestions. Only user-spawned agents represent intentional delegation decisions by the human or parent agent.

4.3 Temporal evolution (RQ3)

The network evolves over 49 daily snapshots spanning 101 calendar days (Fig. 5). Session creation shows two phases: a ramp-up period (late November 2025 through late January 2026, with sporadic early usage) followed by intensive sustained use (February 2026 onward, averaging approximately 80 sessions per active day).

Densification. The relationship between connected nodes and edges follows a power law with high fidelity (Fig. 6):

$$\gamma = 1.707, \quad R^2 = 0.988, \quad p < 10^{-45} \quad (4)$$

This super-linear exponent is substantially stronger than the conversational baseline ($\gamma = 1.405$, $R^2 = 0.993$). For context, Leskovec et al. [2007] report $\gamma = 1.69$ for arXiv citation networks, $\gamma = 1.26$ for patent networks, and $\gamma = 1.18$ for autonomous systems. The agentic knowledge network densifies comparably to citation networks.

Table 4 places these results in the context of established densification exponents.

Giant component evolution. The giant component fraction rises rapidly during the ramp-up period, reaching approximately 60% by early February 2026, and stabilizes near 62.5% thereafter. The stabilization is notably faster than in the conversational network, where the giant component fraction fluctuated over many months before settling. This likely reflects the higher temporal density

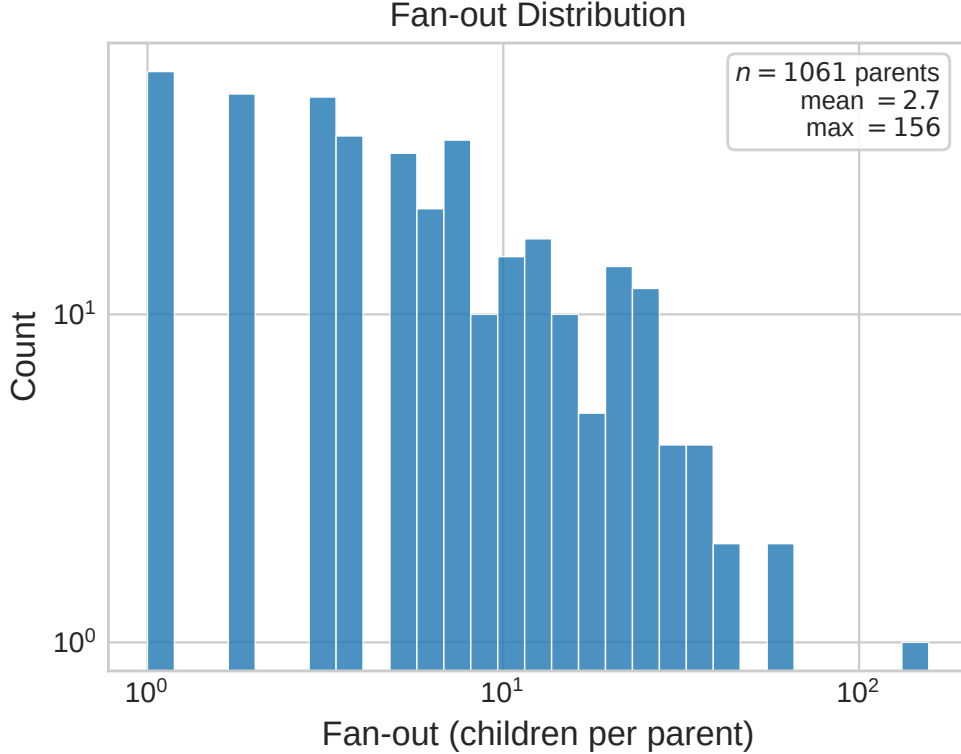


Figure 3: Fan-out distribution (log-log). Most parents spawn 0–3 children, but a heavy tail extends to 156.

of agentic sessions: the semantic space fills in within weeks rather than years.

Fan-out dynamics. Mean daily fan-out (children per parent) peaks at approximately 4.3 in late January 2026 during a period of intensive multi-agent development work, then stabilizes around 2.7 as usage patterns become more routine.

Preferential attachment. Following the methodology of Jeong et al. [2003], we estimate the preferential attachment exponent by recording the degree of each endpoint when new edges appear across successive snapshots and fitting a power law $\Pi(k) \propto k^\beta$. The result is $\beta = -0.082$ ($R^2 = 0.009$, $p = 0.107$), indicating *no evidence for preferential attachment* in the agentic network. This contrasts sharply with the conversational network ($\beta = 0.763$, Paper 2), where new conversations preferentially connected to already-popular topics. In the agentic setting, new semantic connections form independently of existing degree, consistent with project-driven work where sessions connect to whatever the current task requires rather than gravitating toward semantic hubs.

4.4 Multi-layer analysis (RQ4)

Table 5 summarizes the multi-layer metrics.

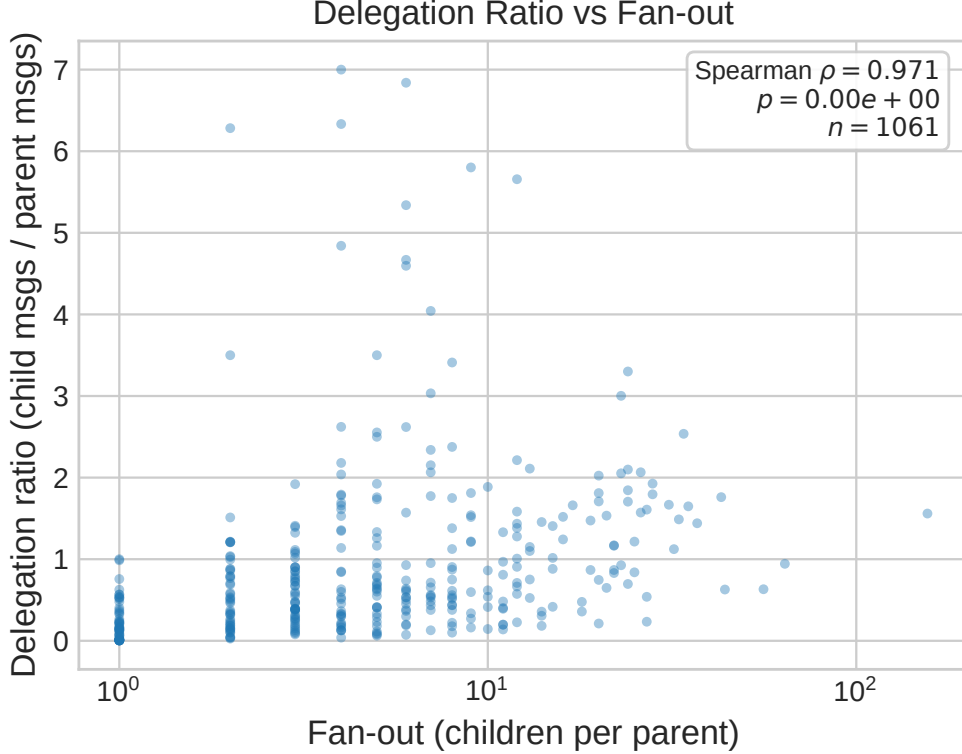


Figure 4: Delegation ratio versus fan-out. Higher fan-out parents tend to have lower per-child delegation ratios, suggesting that high-delegation sessions decompose into many small subtasks.

Inter-layer degree correlation. The Spearman rank correlation between semantic degree and delegation degree is $\rho = 0.115$ ($p < 10^{-13}$, $n = 3,945$; Fig. 8). This is statistically significant—the p -value reflects the large sample size—but substantively weak. Knowing a session’s semantic centrality (how many other sessions it resembles topically) provides minimal information about its delegation behavior (how many children it spawned or was spawned by). The two layers encode largely independent structure.

Participation coefficient. The mean multiplex participation coefficient is $P = 0.142$ (Fig. 7), with a maximum of 0.5 (the theoretical maximum for two layers). Most sessions are specialized to one layer: they have many semantic connections but few delegation connections (leaf nodes in the hierarchy), or they are delegation hubs with moderate semantic connectivity. Only a small fraction of sessions achieve balanced participation across both layers (high P), representing semantically well-connected sessions that are also heavily involved in delegation.

Community NMI. The Normalized Mutual Information between semantic communities (Louvain) and delegation communities (connected components of the directed tree structure) is $\text{NMI} = 0.581$. This moderate value indicates partial overlap: parent–child groups tend to be in the same semantic neighborhood (unsurprisingly, since delegated subtasks often relate to the parent’s topic), but the

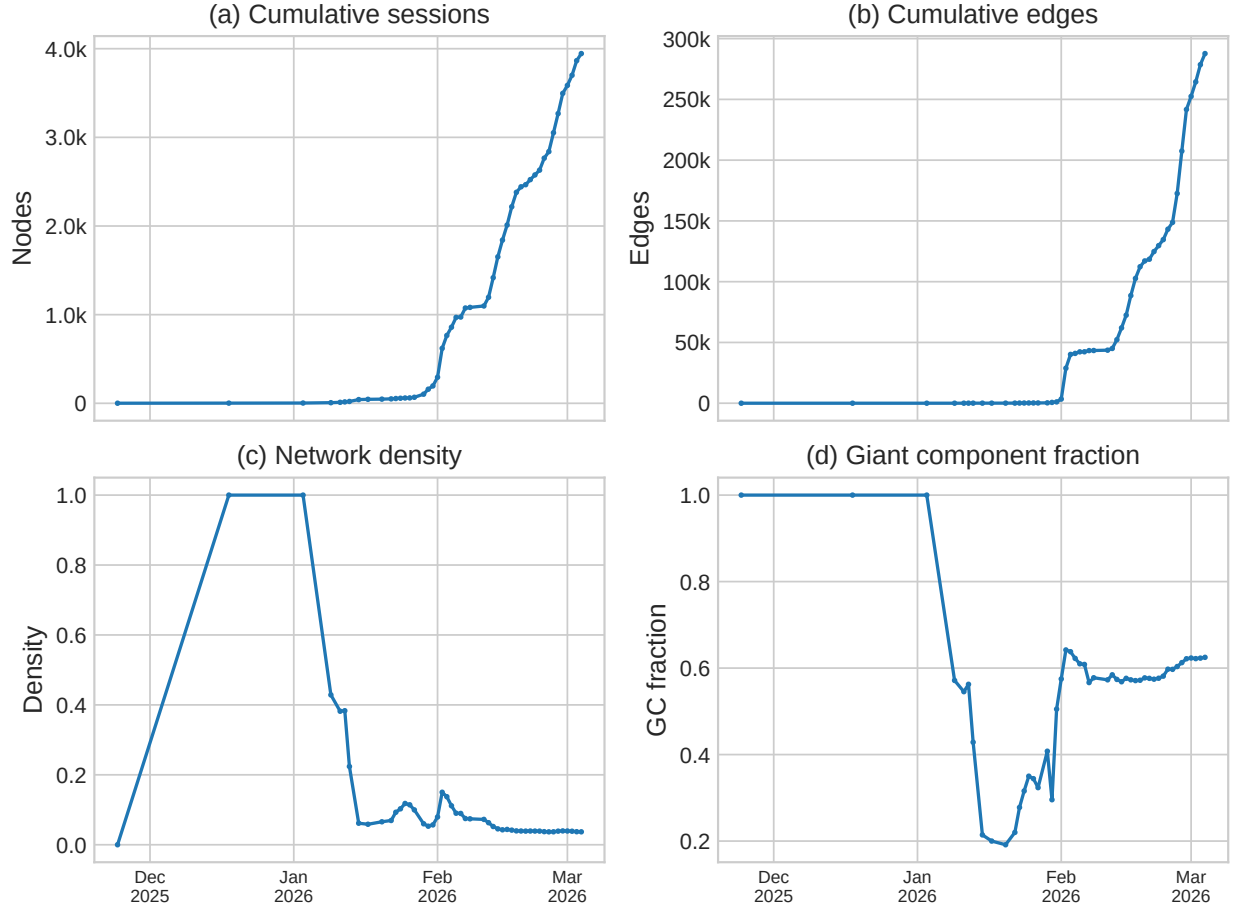


Figure 5: Temporal evolution of network properties across 49 daily snapshots. (a) Cumulative sessions grow rapidly during February 2026. (b) Edges grow super-linearly. (c) Density rises then stabilizes. (d) Giant component fraction reaches 62.5%.

correspondence is far from perfect. Semantic communities group topically similar sessions that may span many different delegation trees, while delegation components group hierarchically related sessions that may span different topics.

4.5 Null model validation (RQ5)

To confirm that the observed network properties reflect genuine structure rather than artifacts of the degree sequence or node count, we test all key metrics against appropriate null models.

Configuration model baseline. We generate 20 random graphs preserving the observed degree sequence via the configuration model [Newman, 2003] and compare observed metrics against the null ensemble (Table 6).

All four metrics deviate massively from the configuration model null ($z > 69$ in every case). The observed clustering coefficient is nearly twice the null expectation, transitivity is $2.6\times$ higher, and

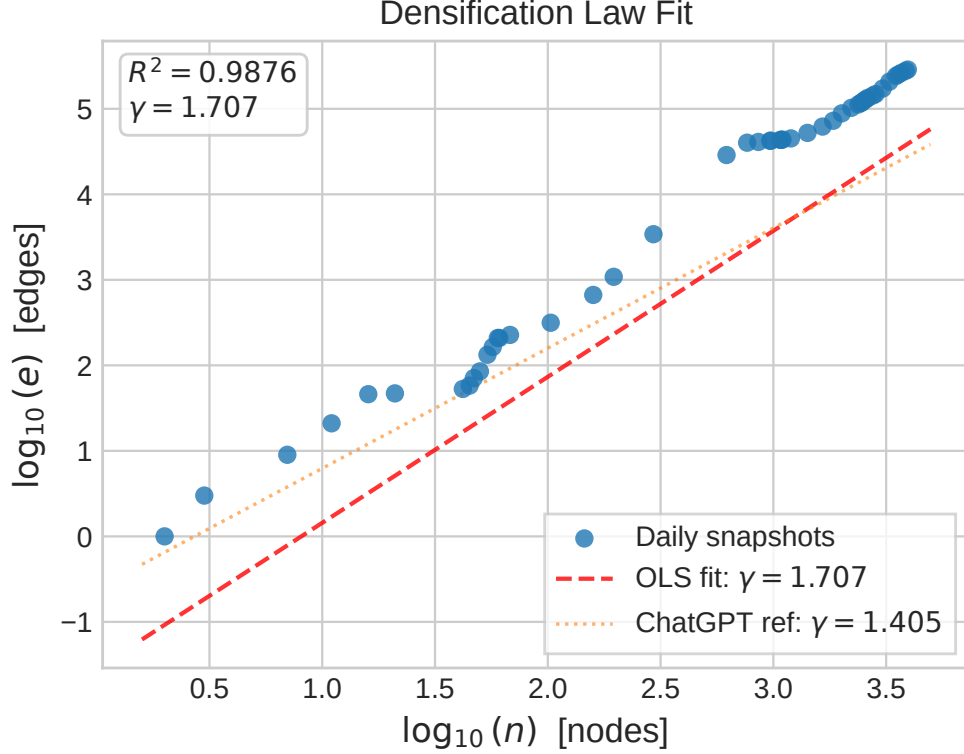


Figure 6: Densification law fit: $\log e$ vs. $\log n$ for daily snapshots ($\gamma = 1.707$, $R^2 = 0.988$). The ChatGPT reference line ($\gamma = 1.405$) is shown for comparison.

modularity is $9.5\times$ the null value. The assortativity reversal is especially striking: configuration model graphs are weakly disassortative (-0.032), while the observed network is strongly assortative ($+0.901$). These results confirm that the semantic network’s community structure, transitivity, and degree correlations are genuine properties of the knowledge organization, not artifacts of the degree distribution.

Permutation test: inter-layer correlation. Permuting node labels in the delegation layer across 1,000 permutations yields a null distribution of Spearman ρ centered at -0.001 (std = 0.016). The observed $\rho = 0.115$ exceeds all 1,000 permuted values ($p < 0.001$), confirming that the inter-layer degree correlation—though substantively weak—is statistically genuine. Semantic hubs and delegation hubs overlap more than chance allows.

Permutation test: community NMI. Permuting community labels across 1,000 permutations yields a null NMI distribution centered at 0.504 (std = 0.002). The observed NMI = 0.581 exceeds all permuted values ($p < 0.001$). The alignment between semantic communities and delegation components is significantly stronger than expected under random assignment, confirming that delegation structure partially reflects semantic organization.

Table 4: Densification exponents across network types.

Network	Exponent γ	Source
arXiv citations	1.69	Leskovec et al. [2007]
Agentic knowledge (this work)	1.707	—
Conversational knowledge (Paper 2)	1.405	Towell and Matta [2026b]
Patent citations	1.26	Leskovec et al. [2007]
Autonomous systems	1.18	Leskovec et al. [2007]

Table 5: Multi-layer analysis summary.

Metric	Value
Inter-layer degree correlation (ρ)	0.115
p -value	$< 10^{-13}$
Shared nodes	3,945
Participation coefficient (mean)	0.142
Participation coefficient (max)	0.500
Community NMI	0.581

5 Discussion

5.1 Why densification intensifies in agentic mode

The increase in densification exponent from $\gamma = 1.405$ (conversational) to $\gamma = 1.707$ (agentic) has a natural interpretation rooted in the difference between conversation and tool use. In conversational AI, a session typically addresses a single topic: the user asks about machine learning, and the conversation stays in the machine learning domain. The resulting embedding occupies a specific region of semantic space.

In agentic AI, a single session may read documentation (one domain), edit configuration files (another domain), run tests (yet another), and fix bugs across multiple modules. The embedding of such a session reflects all these activities, placing it in a broader region of semantic space where it intersects with sessions from many different projects. The result is more cross-domain connections per node added, which drives the higher densification exponent.

The exponent $\gamma = 1.707$ places the agentic knowledge network alongside arXiv citation networks ($\gamma = 1.69$) in the [Leskovec et al. \[2007\]](#) taxonomy. This is a suggestive parallel: citation networks densify because papers increasingly reference a growing body of prior work, while agentic sessions densify because tool use creates increasingly cross-domain semantic footprints. The mechanisms differ, but the macroscopic growth laws converge.

5.2 From topics to projects: why modularity decreases

The drop in modularity from 0.750 to 0.391 does not indicate a loss of structure; it indicates a different *kind* of structure. The ChatGPT network organized into 15 well-separated knowledge

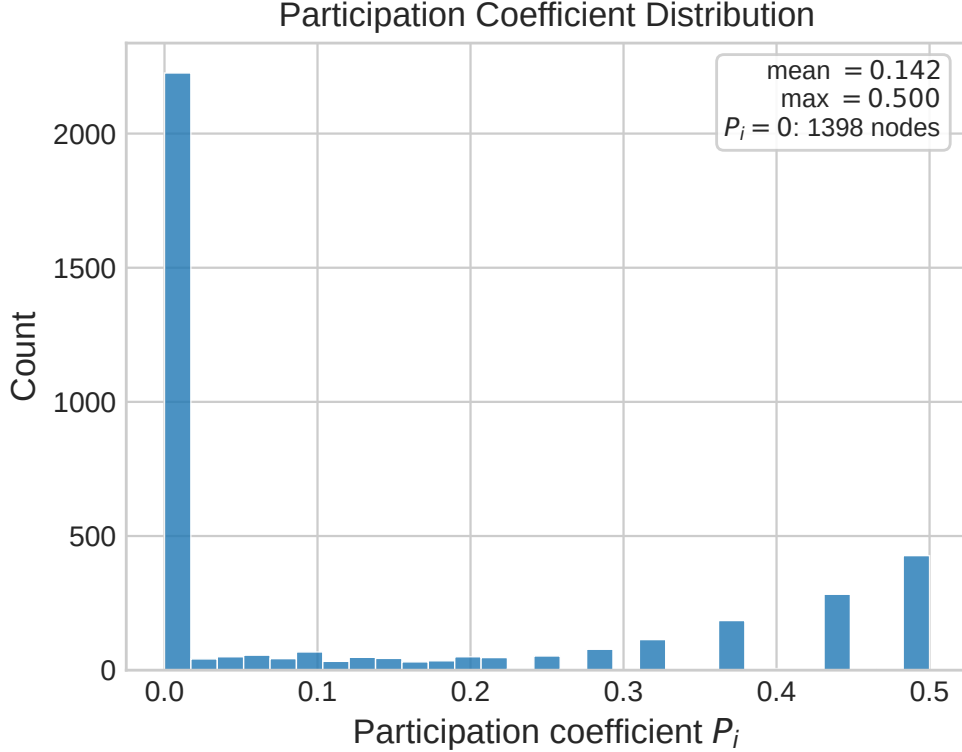


Figure 7: Distribution of multiplex participation coefficients. Most sessions specialize in one layer ($P_i \approx 0$); few achieve balanced participation across both semantic and delegation layers.

communities—machine learning, statistics, philosophy, programming—each internally dense and externally sparse. These reflect topical coherence: conversations about statistics cluster with other conversations about statistics.

The Claude Code network organizes into 1,084 smaller, less distinct clusters. This reflects project-driven rather than topic-driven organization. In agentic work, the organizing unit is the project or task, not the topic. A project to “build a web scraper” involves HTTP requests, HTML parsing, data storage, and error handling—four traditional topic categories in one task. Community detection algorithms optimized for topical separation produce many small clusters because the topics are intrinsically mixed within sessions.

Lower modularity in this context is not a deficiency of the agentic network; it is a signature of how tool-mediated work organizes knowledge differently from conversation. Community detection approaches designed for overlapping or hierarchical communities [Palla et al., 2007] may be better suited to this structure than Louvain’s modularity optimization.

5.3 The complementarity of semantic and delegation layers

The central finding of this paper is the weak inter-layer degree correlation ($\rho = 0.115$) and moderate community NMI (0.581). These results mean that the semantic and delegation layers capture

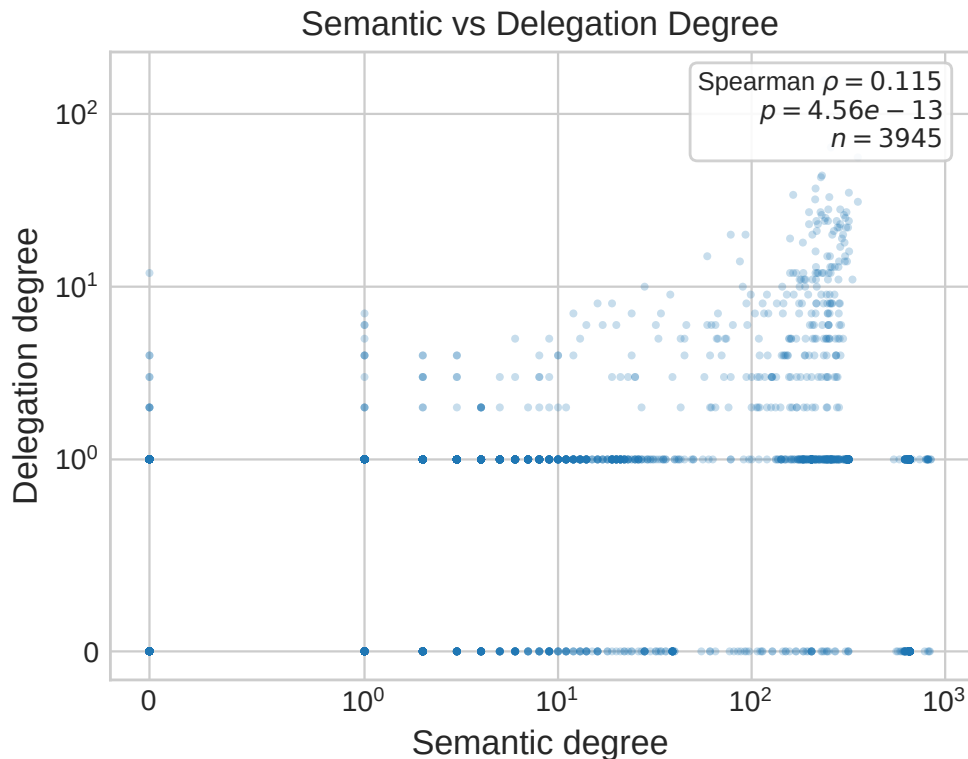


Figure 8: Semantic degree vs. delegation degree for each session. The weak correlation ($\rho = 0.115$) confirms that the two layers capture complementary structure.

genuinely complementary dimensions of cognitive activity. The correlation is nonzero—semantic hubs and delegation hubs overlap more than chance allows ($p < 0.001$)—but substantively weak: knowing a session’s semantic degree explains barely 1% of the variance in its delegation degree.

This complementarity has a clear interpretation through the lens of distributed cognition [Hutchins, 1995]. The semantic layer captures *what knowledge is being explored*: topical relationships between sessions regardless of delegation. The delegation layer captures *how cognitive labor is distributed*: which sessions spawned others, how work was decomposed, and what structural roles different agent types play. These are fundamentally different dimensions of intellectual activity, and the empirical weak coupling confirms that neither can be recovered from the other.

This has methodological implications for our prior work. Papers 1–2 analyzed conversational AI archives using only semantic similarity, appropriate because conversational AI has no delegation structure. But agentic AI *does* have delegation structure, and our results show that semantic analysis alone misses it. Single-layer analysis of agentic archives is incomplete in a way that single-layer analysis of conversational archives is not.

The observed complementarity is consistent with findings in other multiplex systems. Szell et al. [2010] found that social, economic, and enmity layers in an online game encode largely distinct structure. Bullmore and Sporns [2009] found that structural and functional layers in brain networks capture different aspects of neural organization. The principle that different relationship types in

Table 6: Configuration model baseline comparison ($n = 20$ samples). All observed values exceeded every null sample.

Metric	Observed	Null mean $\pm$ std	z -score
Clustering	0.481	0.243 ± 0.003	69.2
Transitivity	0.957	0.364 ± 0.000	1,338
Assortativity	0.901	-0.032 ± 0.002	411
Modularity	0.391	0.041 ± 0.001	406

multiplex systems encode complementary rather than redundant information appears to extend to human–AI cognitive traces.

5.4 The densification paradox revisited

Paper 2 noted that densification in semantic similarity networks reflects the *progressive revelation* of latent structure rather than *active tie formation*: all potential edges exist the moment both endpoint sessions are created, and the densification we measure describes how the pre-existing similarity structure fills in over time. This caveat applies with equal force here. The agentic network starts denser and adds proportionally more edges, but the edges are still determined by a pre-computed similarity matrix.

However, the *comparison* between conversational and agentic densification exponents is valid precisely because both networks use the same threshold, embedding model, and human subject. The difference in γ (1.707 vs. 1.405) reflects genuine structural differences in how agentic vs. conversational content populates semantic space, not an artifact of the method.

The delegation layer, by contrast, represents actual tie formation: a parent session actively spawns a child session. The delegation network involves no thresholding, no embedding, and no post-hoc computation. Its structure is ground truth.

5.5 Implications for agentic AI design

Our findings suggest several design implications for agentic AI systems and the tools that manage their outputs.

First, *delegation is not just task decomposition*. The weak coupling of semantic and delegation layers shows that delegation patterns reflect more than dividing a task into subtopics. Compact agents and prompt suggestion agents play structural roles distinct from user-spawned agents, and the delegation hierarchy encodes organizational information absent from the semantic content.

Second, *extreme delegation warrants monitoring*. The heavy-tailed fan-out distribution, with one session spawning 156 children, suggests that some sessions attempt extreme delegation. Whether this is efficient (genuine parallelism of independent subtasks) or pathological (thrashing, redundant retries) is an empirical question that delegation network analysis could help answer.

Third, *session management should be multiplex-aware*. Conversation management tools that

organize by topic alone would miss the delegation structure connecting related sessions across different parts of the hierarchy. Tools that show only the parent–child tree would miss semantic connections between sessions in different delegation trees. Effective organization of agentic AI archives requires a multiplex view that surfaces both layers.

5.6 Limitations

Several limitations constrain the generalizability of our findings.

Single subject. This remains a case study of one user’s AI session archive. The specific parameter values (γ , ρ , NMI, modularity) may differ for other users. Multi-user studies are needed to establish which patterns generalize.

Short observation period. The 101-day observation window is substantially shorter than the 29-month window in Paper 2. Long-term dynamics—community lifecycle patterns, stabilization trajectories—cannot be assessed. The densification comparison should be interpreted as capturing different growth regimes (early intensive use vs. extended moderate use) rather than as a controlled temporal comparison.

Single embedding configuration. We use text-only preprocessing with no ablation study. Stripping tool-use and tool-result blocks discards information that may be semantically important for agentic sessions. Code-aware embeddings, or embeddings that incorporate tool-use structure, might produce different semantic layer properties. This is a priority for future work.

Single threshold. The similarity threshold $\theta = 0.9$ was validated for conversational data [Towell and Matta, 2025] but not independently for agentic data. The phase transition behavior may differ when sessions contain mixed content modalities and vary more dramatically in length (from a brief subagent lookup to a 1,057-message development session).

Subagent heterogeneity. Including compact and prompt suggestion agents inflates the node count and may distort network metrics. These sessions are infrastructure, not intellectual work. Future analysis should stratify by agent type, comparing the network with all sessions versus user-spawned agents only.

Limited null models. Configuration model baselines and permutation tests confirm that semantic network metrics and inter-layer correlations are statistically genuine (Section 4.5). However, we lack a temporal null model for the densification exponent—testing whether $\gamma = 1.707$ exceeds what random session arrival with the observed similarity distribution would produce remains future work.

Depth-2 hierarchy. Claude Code’s current architecture limits delegation to parent–child (no grandchildren). Future agentic systems with deeper delegation hierarchies may produce richer tree structures that enable analysis of recursive delegation patterns.

6 Conclusion

We have extended the cognitive MRI methodology from single-layer analysis of conversational AI to multiplex analysis of agentic AI, constructing and analyzing the first two-layer network

representation of an agentic AI session archive. The analysis of 3,945 Claude Code sessions from the same human subject studied in our prior work reveals three principal findings.

First, agentic work intensifies densification ($\gamma = 1.707$ vs. 1.405), placing the agentic knowledge network alongside citation networks in the landscape of densifying complex systems. Tool use creates more cross-domain semantic connections than conversation, consistent with the multi-topic nature of agentic coding sessions.

Second, the structural character of knowledge organization shifts from topic-driven to project-driven: modularity decreases (0.391 vs. 0.750), assortativity reverses (+0.90 vs. -0.13), and the network fragments into many small communities rather than a few large ones. Agentic work is organized by tasks, not topics.

Third, the semantic similarity and delegation hierarchy layers are weakly coupled ($\rho = 0.115$, NMI = 0.581). This is the most consequential finding: it means that single-layer analysis—whether semantic or structural—systematically misses the complementary dimension that the other layer encodes. The multiplex view is not merely richer; it captures structure invisible to either layer alone.

These findings suggest that as AI transitions from passive interlocutor to autonomous agent, the cognitive traces it produces become structurally more complex, demanding analytical frameworks that can represent multiple simultaneous relationship types. The multiplex network perspective, well-established in social and biological network analysis, proves essential for understanding agentic AI archives. Future work should extend this analysis to multiple users, incorporate code-aware embeddings, develop temporal null models for densification, and track how the multiplex structure evolves as agentic AI tools continue to mature and delegation hierarchies deepen.

Data availability

The analysis pipeline, session embeddings (with message text removed for privacy), edge lists, and all derived data are available at <https://github.com/queelius/cognitive-mri-conversations> [Towell and Matta, 2026a]. The original analysis pipeline for Papers 1–2 is available at <https://github.com/queelius/chatgpt-complex-net> [Towell, 2025]. Raw session content is not shared to protect the privacy of the human participant.

Author contributions

Alexander Towell: Conceptualization, Methodology, Software, Formal Analysis, Data Curation, Writing – Original Draft, Visualization. **John Matta:** Supervision, Writing – Review & Editing.

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Competing interests

The authors declare no competing interests.

Ethics statement

This study analyzes one author’s own Claude Code session archive. No human subjects were recruited, and no personally identifiable information about third parties is included in the dataset or analysis. The session data was generated through the author’s routine use of a commercially available AI coding assistant.

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